HUMAIRA ANZUM NEHA

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Technical Summary

- Research experience in human-centered AI and accessibility for people with visual impairments and people who stutter.
- Research experience in qualitative HCI studies, including workshops, semi-structured interviews, and thematic analysis.
- Research experience in computer vision for object detection, currency recognition, and face recognition using deep learning models e.g., YOLOv5, ResNet-50.
- Research experience in assistive wearables with distance and motion sensors for fall detection and obstacle avoidance.
- Developed a mobile application integrated with a deep learning model for non-invasive glucose and hemoglobin estimation through fingertip video processing.
- Hands-on experience in collecting and annotating custom image datasets for supervised model training.
- Hands-on experience in deep learning frameworks and libraries e.g., TensorFlow, Keras, PyTorch, Hugging Face.
- Developed a model agnostic training-free method for long video understanding with vision-language models.
- Developed a multimodal fusion pipeline for audio and text using graph neural networks and BERT.
- Hands-on experience in deploying deep learning models on edge devices such as Raspberry Pi.

Education

M.Sc in CSE at BRAC University **Jan. 2026 – Present**
Completed Credits: 12 *Dhaka, Bangladesh*
- CGPA: 4.00/4.00

B.Sc. in CSE at KUET **Feb. 2019 – Mar. 2024**
Thesis: Real-Time Object, Currency, and Face Recognition for Blind and Low-Vision People *Khulna, Bangladesh*
- CGPA: 3.87/4.00, ranked 3rd in the department

Skills

Programming Languages: Python, C, C++, Java, Kotlin, Javascript, Bash, and SQL.
Deep Learning Libraries: TensorFlow, Keras, and PyTorch, Hugging Face
AI Tools: OpenAI Codex and Claude.
Developer Tools: Anaconda, Jupyter, Arduino, Raspberry Pi , Git, MS Visual C++, and VS Code.
Database: MySQL, Oracle, Firebase, MongoDB
Mobile Apps Development: Android Studio.
Operating System: Linux, MacOs, and Windows.
Research Aptitude: Human-Centered AI, Accessibility, Computer Vision, and Multimodal AI
Adaptive Skills: Mentoring and Leadership, Collaborative Research, Community-Engaged Research, Academic Writing, Complex Problem Solving, and Rapid Skill Acquisition.

Experience

Lecturer **Aug 2024 – Present**
Computer Science and Engineering, United International University *Dhaka, Bangladesh*
- Leading a project to study the reliability of vision-language models in reading and reasoning across diverse expiry date formats for blind and low-vision users, along with strategies to improve their performance.
- Taught undergraduate courses including Data Structures and Algorithms, Database Management Systems, Data Mining, Object Oriented Programming, and Computer Architecture.
- Supervised undergraduate thesis projects, mentored student teams for competitions, and judged departmental project shows.

Projects

Video Creation and Editing among People Who Stutter | *Semi-Structured Interview* **May. 2026 - Sep. 2026**
- Conducted an onboarding workshop and semi-structured interviews with eight people who stutter in Bangladesh to understand how they engage in video content creation to represent themselves and their community.
- Analyzed the interview data using reflexive thematic analysis to uncover how video context and audience shape participants' editing choices and what support they need from AI-assisted editing tools. Manuscript Under Review

Query-Guided Visual Prompting for Long Video Understanding | *Python, PyTorch, VLMs* **Feb. 2026 - Present**

- Designed a frame scoring method using a lightweight vision-language model (CLIP) to select question-relevant frames and arrange them into asymmetric grids that give more space to more relevant frames and used them as visual prompts for vision-language models without any fine-tuning.
- Improved pair accuracy on the MF2 benchmark from 16.9% to 23.5% over the baseline using Qwen2.5-VL-7B.

GNN–BERT Fusion for Understanding Music Context | *PyTorch, GNN, BERT* **Jan. 2026 - May. 2026**

- Leveraged graph neural network models to build an audio graph and fused them into a pre-trained large language model (BERT) for textual feature extraction and class tagging.
- Cross-attention fusion raised micro-F1 on MagnaTagATune from 0.38 (BERT-only) to 0.52.

Object and Currency Recognition for the Visually Impaired | *PyTorch* **Jul. 2023 - Jun. 2024**

- Trained real-time object detection models (YOLOv5) on a custom annotated dataset to detect objects and classify Bangladeshi banknotes and deployed them on an edge device (Raspberry Pi).
- Leveraged ResNet-50 for face recognition to help users identify familiar people.
- Achieved 98.2% mAP for objects, 99.7% top-1 accuracy for currency, and 97.6% accuracy for face recognition.

Assistive Glasses for Navigation, Fall Detection, and Reading | *Raspberry Pi, Pytesseract* **Jun. 2023 - May. 2024**

- Integrated distance sensors (Time-of-Flight) with a motion sensor (accelerometer) on an edge device (Raspberry Pi) to provide users with real-time obstacle alerts and send fall alerts to their caregivers.
- Implemented an optical character recognition (OCR) and text-to-speech pipeline that reads printed text aloud to users with 93.03% character accuracy.

Mobile Application to Measure Blood Component Levels | *Kotlin, Python, OpenCV* **Dec. 2022 - May. 2023**

- Developed a mobile application that records a 15-second fingertip video and extracts photoplethysmogram (PPG) signals from it to estimate glucose and hemoglobin levels using a deep learning model.
- Calibrated camera settings for consistent video quality across lighting conditions and devices.

Abalone Board Game with AI Opponent | *Python, Pygame* **Nov. 2022 - Feb. 2023**

- Developed a board game to play with an AI opponent using Minimax search and alpha–beta pruning.

Relevant Coursework

- Advanced Natural Language Processing
- Deep Learning for Computer Vision
- Research Methodology and Seminar (HCI-Focused)
- Neural Networks and Fuzzy Systems
- Fairness, Ethics and Explainability in AI
- Computer and Network Security
- Data Mining
- Mobile Computing
- Artificial Intelligence
- Embedded Systems
- Image Processing
- Applied Statistics

Publications

- **H. Neha**, M. S. Sadi, M. R. Islam and S. Sakib, "Automated Object Identification for the Visually Impaired," 2024 27th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2024, pp. 3584-3589, doi: 10.1109/ICCIT64611.2024.11022362.
- S. Sakib, M. S. Sadi, M. S. Arefin, M. R. Islam and **H. Neha**, "An Assistive Device for the Visually Impaired for Enhanced Navigation and Reading," 2024 13th International Conference on Electrical and Computer Engineering (ICECE), Dhaka, Bangladesh, 2024, pp. 767-772, doi: 10.1109/ICECE64886.2024.11024586.
- **H. Neha**, S. Sakib, F. Sadaf and S. M. Taslim Uddin Raju, "Mobile Application to Collect Data and Measure Blood Component Level in a Non-Invasive Way," 2023 26th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2023, pp. 1-6, doi: 10.1109/ICCIT60459.2023.10441186.

Achievements

- **Top 3 CGPA** among 120 students in the CSE department, Khulna University of Engineering & Technology (KUET).
- **Dean's List Award** (3 times) for academic excellence, KUET.
- **Top-10 Finalist**, UNDP Women's Digital Innovation Hackathon 2021, for a community app for parents of autistic children.

Standardized Test Scores

TOEFL iBT **Oct. 2026**

Overall: 5.0/6.0 (new 1–6 scale)

- Reading: 5.0, Listening: 5.5, Speaking: 5.0, Writing: 5.0